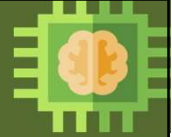


Elective Course

Course Code: CS4103

Autumn 2026-27

**Lecture #01**

Artificial Intelligence for Data Science

Module-1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE (AI)

AI Introduction, Brief history

Course Details

Course Instructor:**Dr. Monidipa Das**

Assistant Professor

Department of Computational and Data Sciences

Indian Institute of Science Education and Research Kolkata, India 741246



AI Introduction

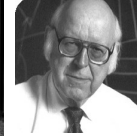
- What is AI?
- Definition of AI
- Typical AI Problems
- Key Topics of AI
- Foundations of AI
- Applications of AI
- Ethical and Societal Concerns with AI
- History of AI

What is AI?



- **Artificial Intelligence**

- Term coined by **J. McCarthy** in 1956
- Concerned with the design of intelligence in an artificial device



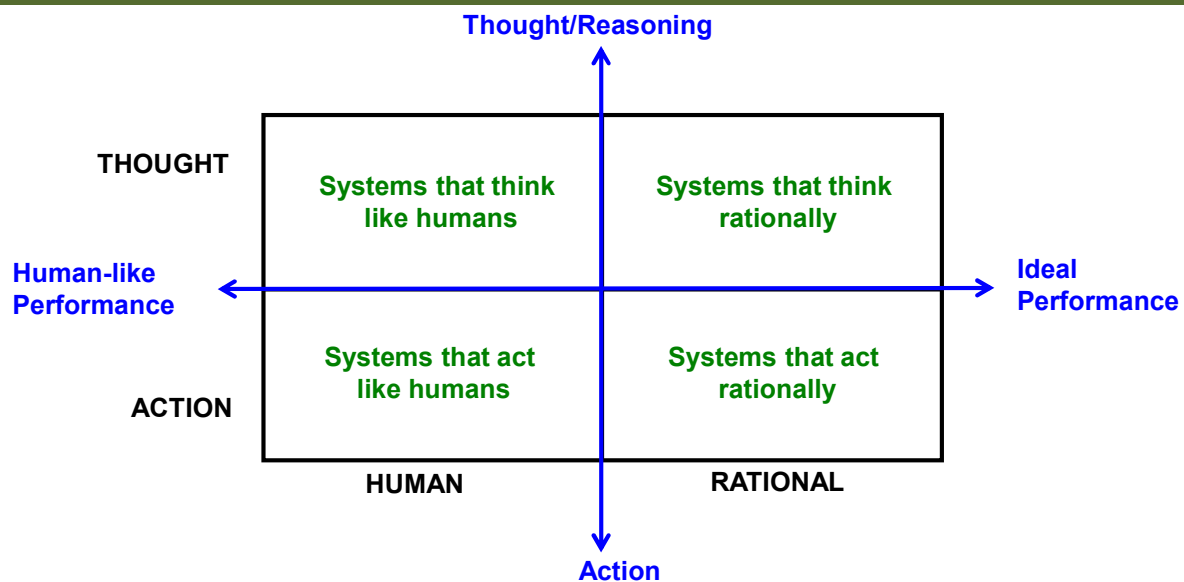
- **What is Intelligence?**

Intelligence is defined by some general characteristics:

- **Perception:** Manipulation and interpretation of data.
- **Action:** Control and use of effectors to accomplish a variety of tasks.
- **Reasoning:** Deductive (logical) inference, inductive inference.
- **Learning:** Adapting behavior to better cope with changing environments, discovery of patterns, learning to reason, plan, and act.
- **Communication** with other intelligent agents including humans using signals, signs, icons, etc.
- **Planning:** Formulation of plans - sequences or agenda of actions to accomplish externally or internally determined goals.

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Definitions of AI



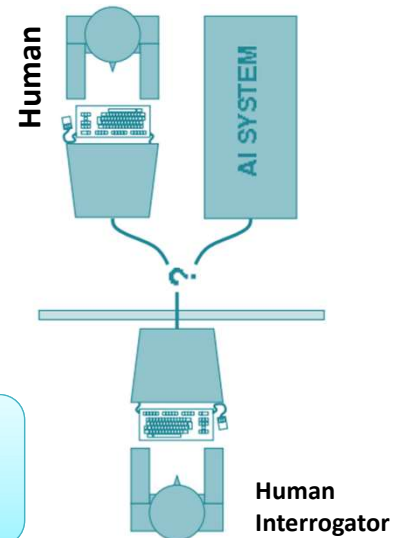
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Systems that act like humans: Turing Test



- You enter a room which has a computer terminal. You have a fixed period of time to type what you want into the terminal, and study the replies. At the other end of the line is either a human being or a computer system.
- If it is a computer system, and at the end of the period you cannot reliably determine whether it is a system or a human, then the system is deemed to be intelligent.

- *"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990)*
- *"The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)*



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Systems that think like humans: Cognitive Modeling



- **Cognitive Science**

There are three ways to do this:

- through introspection—trying to catch our own thoughts as they go by;
- through psychological experiments—observing a person in action;
- through brain imaging—observing the brain in action.

- *"The exciting new effort to make computers think . . . machines with minds, in the full and literal sense." (Haugeland, 1985)*
- *"[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)*

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Systems that think 'rationally': Laws of Thought



- Humans are not always 'rational'
- Logic can't express everything
- Logical approach is often not feasible in practice

- *"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985)*
- *"The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)*

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Systems that act rationally: Rational Agents



- **Rational Agents:**
 - They operate autonomously,
 - perceive their environment,
 - persist over a prolonged time period,
 - adapt to change
 - create and pursue goals

- *"Computational Intelligence is the study of the design of intelligent agents." (Poole et al., 1998)*
- *"AI ...is concerned with intelligent behavior in artifacts." (Nilsson, 1998)*

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Typical AI Problems



- ❑ The intelligent agents or entities need to be able to do both mundane tasks and expert tasks

❑ Mundane Tasks

- Planning Route
- Recognizing (vision)
- Communicating (though natural language : Understanding, Generation, Translation)
- Navigating

❑ Expert Task

- Playing Chess
- Medical Diagnosis
- Mathematical Problem Solving

It has been very hard to mechanize many tasks that lot of animals can do very easily.

“I went to the grocery store, I saw the milk on the shelf and I bought it.”

What did I buy?

- The milk?
- The shelf?
- The store?

An awful lot of reasoning abilities and knowledge of the world is needed to answer simple questions like this one

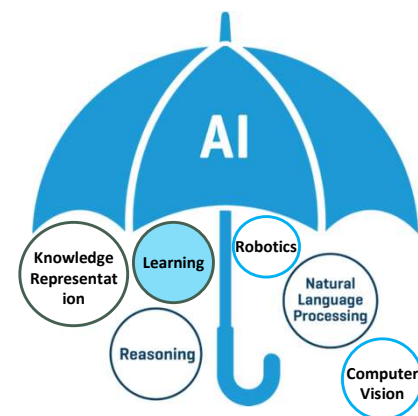
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Fundamental Topics in AI



Artificial intelligence can be considered under a number of headings:

- Search Techniques
- Constraint Satisfaction Problems
- Genetic Algorithms
- Representing Knowledge and Reasoning with it
- Probabilistic Reasoning
- Learning
- Expert Systems
- Planning
- Interacting with the Environment
- Natural language processing
- Fuzzy Systems

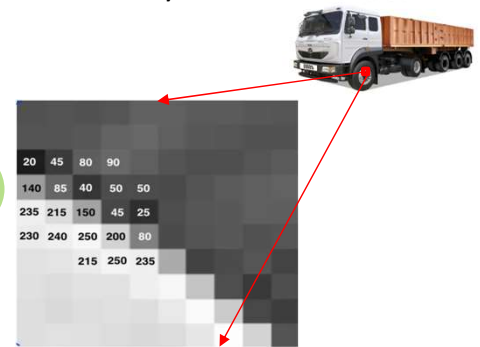
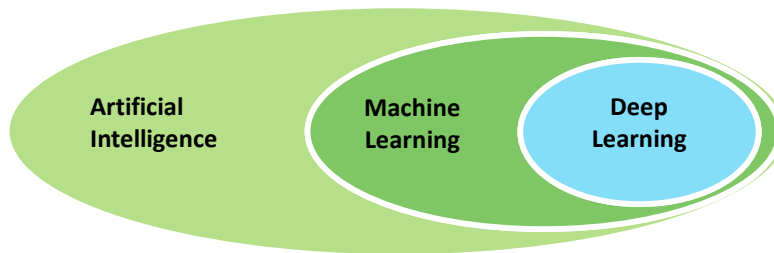


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Fundamental Topics in AI: Learning



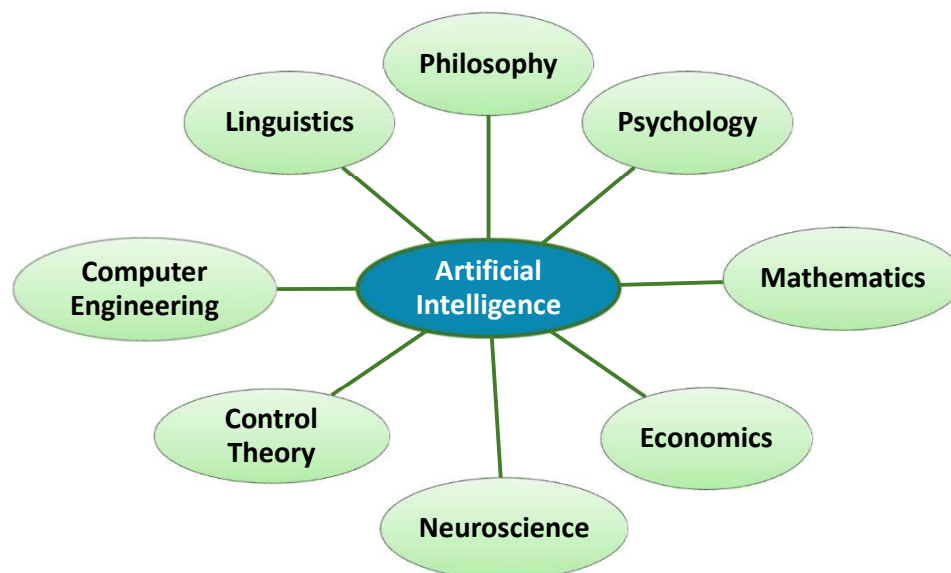
- **Machine Learning:** “field of study that gives computers the ability to learn without explicitly being programmed”



- **Deep Learning:** “computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction.”

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Foundations of AI



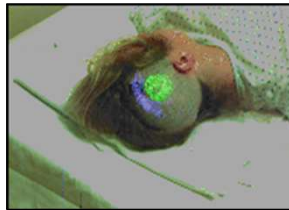
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Applications of AI

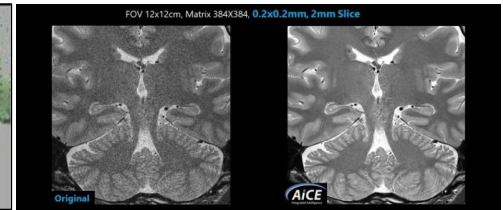


• Medicine

- Image guided surgery
- Image enhancement



Source: http://www.ai.mit.edu/projects/medical-vision/surgery/surgical_navigation.html



Source: <https://healthcare-in-europe.com/en/news/1-5t-mr-system-receives-fda-clearance-for-ai-based-image-reconstruction-technology.html>

• Transportation

- Autonomous car
- Pedestrian Detection



<https://www.mdpi.com/2076-2489/14/2/123>



[Lexus RX450h retrofitted as a Google driverless car](https://www.mdpi.com/2076-2489/14/2/123)

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Applications of AI [contd.]



• Games



<https://www.bbc.com/news/technology-35785875>

• Entertainment Agents

• Internet Agents

- Monitors users task,
- seek required information, Learns



<https://electronics.sony.com/>



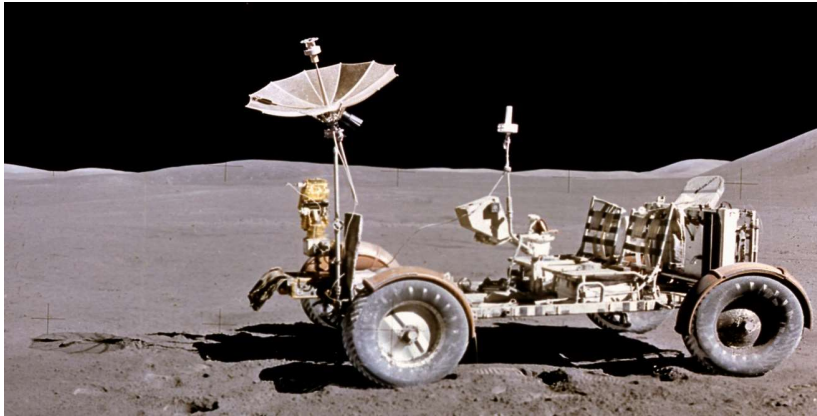
<https://www.genesis-toys.com/>

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Applications of AI [contd.]

15

- Autonomous Planning & Scheduling:
 - Autonomous rovers



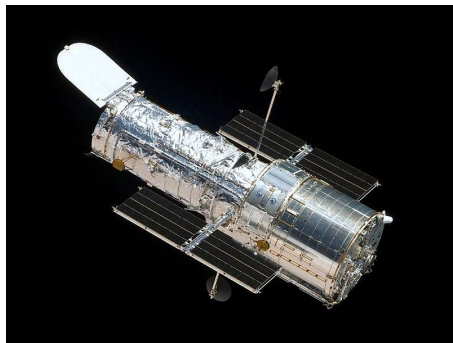
Source: <https://simple.wikipedia.org/>

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Applications of AI [contd.]

16

- Autonomous Planning & Scheduling:
 - Telescope scheduling



Source: <https://en.wikipedia.org/wiki/>



Hubble Space Telescope. Public domain image, NASA

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Applications of AI [contd.]



Other application areas:

- Bioinformatics
- Text classification, document sorting
- Video, image classification
- Music composition, picture drawing
- Natural Language Processing

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Ethical Concerns and Societal Impacts of AI



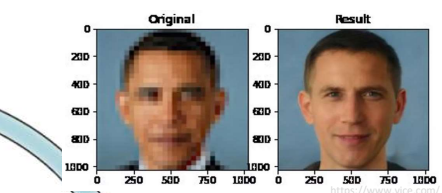
Employment and
Workforce



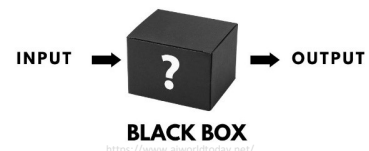
Privacy, Security

Ethical and
societal
implications
of AI

Human
Autonomy



Bias, Fairness,
Transparency,
Accountability



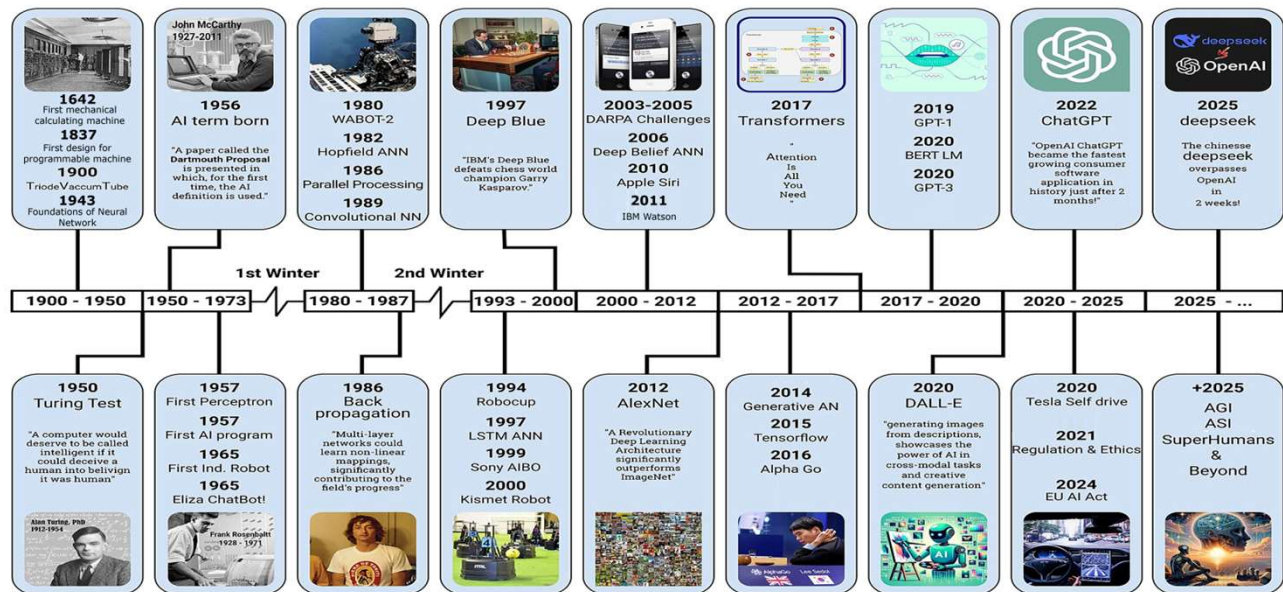
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Brief History of AI

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History of AI



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Course Details

- Course Structure/Syllabus
- Course Materials
- Course Evaluation/Tests

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Course Structure/Syllabus



Syllabus	
Timeline	Topics to be Covered
Week 1	INTRODUCTION TO AI: Artificial Intelligence Introduction, Brief History, Intelligent Agents, Types of agents <i>Python Primer for AI</i>
Week 2-4	PROBLEM SOLVING BY SEARCH: Problem formulation, Concept of state space search Introduction to Uninformed Search Techniques: <i>Breadth First Search, Depth First Search, Depth First Search with Iterative Deepening, Uniform Cost Search</i> Introduction to Informed/Heuristic search techniques: <i>Greedy Best First Search, A* search, Hill Climbing search, Simulated Annealing search</i> Introduction to GA, GA Operations: <i>Selection, Crossover, Mutation</i> <i>Solving N-Queen Problem using GA</i> Adversarial Search: <i>Minimax algorithm, Alpha-Beta Pruning</i> <i>Building a bot to play tic-tac-toe/Building 8-puzzle solver</i>
Week 5	CONSTRAINT SATISFACTION PROBLEM (CSP): Introduction to CSP, Constraint Graph, Binary and Higher order CSP, Backtracking Search, MRV heuristic, Degree heuristic, Least Constraining-value heuristic, Forward Checking, Arc Consistency, Min-Conflicts Algorithm <i>Solving map coloring problem, Solving puzzle</i>

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Course Structure/Syllabus



Syllabus	
Timeline	Topics to be Covered
Week 6-7	INTRODUCTION TO KNOWLEDGE REPRESENTATION AND LOGIC: Propositional Logic (PL) and Reasoning with PL, Concept of Forward and Backward chaining, First Order Logic (FOL) and Reasoning with FOL, Rule-based systems <i>Parsing family tree using logic/Solving puzzle using logic</i>
Week 8	PROBABILISTIC REASONING: Introduction, Probabilistic reasoning with Bayesian Network <i>Disease diagnosis using Bayesian Network (or other domain specific application)</i>
Week 9-13	MACHINE LEARNING (ML): Introduction to the concept of learning, k-Nearest Neighbor (k-NN), Decision tree (DT), Naive-Bayes (NB), Support Vector Machine (SVM), Neural Network (NN) Models <i>Building ML models for Steel Plate Fault detection</i> <i>Building ML models for predicting Aquatic Toxicity (or other domain specific application)</i>
Week 14	Buffer

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Course Materials



- **Text Book:**
 - ❑ **Artificial Intelligence A Modern Approach**, by S. Russell. Norvig, PHI, Third Edition
- **Reference Books:**
 - ❑ **A First Course in Artificial Intelligence** by Deepak Khemani, McGraw Hill Education (India), 2013.
 - ❑ **Artificial Intelligence** by Kevin Knight, Elaine Rich, Third Edition
 - ❑ **Artificial Intelligence: Foundations of Computational Agents** by David L. Poole, Alan K. Mackworth
 - ❑ **Machine Learning** by Mitchell, Tom M., Indian Edition
 - ❑ **Introduction to Machine Learning with Python: A Guide for Data Scientists** by Andreas C. Müller, Sarah Guido

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Evaluation/Tests



- Distribution of Marks:

- ✓ **Mid-Sem Exam:** (converted into **out of 30** marks)
- ✓ **End-Sem Exam:** (converted into **out of 50** marks)
- ✓ **Quiz Tests:** (converted into **out of 20** marks)
Tentative dates: **04-SEP-2026 (Quiz-1)** , **13-NOV-2026 (Quiz-2)**

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Questions?

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